

Vellacombe Discovery Ltd, a Tarnovell company

Vellacombe chemistry service catalogue

Services | January 2025

Date: 21 Jan 2025

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1 Chemistry services at a glance

Vellacombe Discovery Ltd supplies medicinal chemistry from Nottingham. Work starts with a chemical question and ends with a clear statement of what was isolated, what was learned about the route and what remains a proposed structure.

Analogue preparation and Route scout and delivery are the core service names. Impurity isolation and compound-card release support programmes needing a defined reference or a traceable handover.

Table 1. Service scope

Service	Output	Route gate
Analogue preparation	Isolated analogues and compound cards	Identity and recoverable material
Route scout and delivery	Route decision and released material	Isolation supports intended use
Impurity isolation	Assigned reference and interpretation	Feature becomes defined material
Compound-card release	Assessed identity, purity and delivery account	Evidence matches physical material

2 Analogue preparation

Use Analogue preparation for a defined structural comparison: a substituent change, scaffold variation or late-stage diversification evaluated with isolated material. Agree intended structures and the testing purpose before preparation starts.

The deliverable is the isolated analogue set with compound cards. A route attempt can teach useful chemistry without yielding deliverable material; the report states that distinction rather than counting every proposed drawing as a completed analogue.

Ruth Calderwood, Head of Medicinal Chemistry, leads the programme question and route gates. Priorities may change where recovery or identity risk makes a comparison impractical, but changed structures require a recorded decision.

3 Route scout and delivery

Use Route scout and delivery when the route needs an evidence-based decision. Examples include an uncertain transformation, an isolation limiting recovery or a shared intermediate whose handling affects several final analogues.

The scout examines whether the intended structure can be made and isolated for the next operation or client handover. Conversion is useful evidence; the delivery decision depends on isolated material and identity.

Hassan Darwish, Senior Synthetic Chemist, provides route interpretation. The chemistry note identifies the risk, route-gate outcome and next experiment. It is not automatically a manufacturing process package.

4 Isolation and work-up development

A work-up can determine whether a promising transformation becomes a useful route. Crystallisation, precipitation and chromatography are chosen for the material and question rather than treated as interchangeable finishing steps.

Replacing chromatography with crystallisation requires attention to isolated form, solution composition and mother-liquor disposition. A solid's appearance does not establish identity or purity. A further crop requires assessment on the material obtained.

The service can establish a workable isolation within the commissioned scope. Scale, solvent history and incoming-material composition may require another route gate; earlier success does not guarantee unchanged recovery.

5 Impurity isolation

Use impurity isolation when a relevant feature needs to become a defined reference. The question may involve a competing substitution site, heteroaryl side-product or component associated with variable downstream recovery.

Start with the observation making the feature relevant and the source material. An unexplained peak is an observation, not yet an identity. Proposed structures remain hypotheses until supported by isolated-material evidence.

The handover distinguishes reference from desired analogue and states its use. A reference supports comparison and interpretation; it does not establish a numerical incoming-material limit or prove that every later failure has the same cause.

6 Compound-card release

A compound card joins the assigned structure to isolated material, analytical evidence and delivery account. Clara Fenwick, Analytical Release Lead, assesses evidence before material is described as released.

Chromatographic purity is area percent with method context. It is not mass purity and cannot be converted directly into assay concentration. Isolated mass, sampling and dispatched amount remain separate entries.

Unresolved identity or an unsuitable analytical profile places material on hold. A qualified handover requires an explicit decision and a qualification the recipient understands.

7 Inputs required

Provide unambiguous intended structures and describe the programme comparison. State acceptable material forms, downstream use and identity or purity questions that would prevent that use.

For route transfer, supply the existing route, isolation observations and incoming-material identities. For impurity isolation, supply source material and evidence connecting the feature to the process question. Disclose instability or handling sensitivities before shipping.

Table 2. Commissioning inputs

Input	Purpose
Target structures	Avoid site or material-form mismatch
Route and observations	Locate uncertain route gate
Source material identity	Support meaningful impurity comparison
Downstream use	Define appropriate release requirement
Handling qualification	Preserve interpretation at receipt

8 Agree the route gates

Agree the evidence for continuation, replanning or closure before starting. A gate may be isolation of an intermediate, desired alkylation-site assignment or reference availability for an incoming-material question.

Specify the handover separately from experimental ambition. A requested structure is not necessarily accessible; a route proposal is not a delivery promise. The team reports constraints as they become clear.

When a route gate fails, the supported cause and next experiment are useful outputs. Further preparation follows an agreed revised question rather than silent substitution of easier material.

9 Reporting and handover

Reports put delivered-material accounts early and distinguish isolated compounds from proposed structures. Identifiers connect the account to analytical evidence and physical labels. Chemistry conclusions remain separate from biological interpretation.

Andrew May, Sample Management Coordinator, supports delivery-lot reconciliation. The handover identifies material, purpose and qualification. Reference material is labelled so it cannot be confused with a testing analogue.

A report can describe route learning without delivered material. It then states what the scout established and why the delivery gate was not passed.

10 Practical limitations

Related products may be difficult to separate or assign. A clean chromatographic feature may leave positional identity unresolved. High conversion does not guarantee recoverable final material.

Analogue changes can alter solubility, isolation and stability. A shared route is a starting point; each isolated compound needs its own account. Multiple simultaneous structural changes may weaken the intended comparison.

Dates and quantities are agreed for the commissioned scope after inputs and risks are understood. This catalogue gives no universal turnaround, guaranteed yield or fixed price; those decisions depend on the actual chemistry.

11 Choosing the next analogue

The next analogue should answer a specific question using client testing and route learning. A readily prepared structure is not necessarily the most informative comparison.

Ruth Calderwood separates the comparison from synthetic risks. An informative but difficult analogue may justify another route scout; an easy change affecting several properties may leave the question unanswered.

A proposal remains a proposal until material is isolated and released. Retain that language so the testing team knows which structures are physically available.

12 Starting a programme

Bring the question, structures and source material to Ruth Calderwood for scope discussion. Hassan Darwish addresses route uncertainty; Clara Fenwick addresses compound-card release evidence.

The scope records intended material, route gates, release requirements and handover. Where inputs are incomplete, a bounded route scout can resolve uncertainty before a larger analogue preparation is agreed.

Issued on 21 Jan 2025, this catalogue commits to a traceable chemistry outcome: isolated material when the delivery gate passes, clear route learning and a practical next decision where further work is needed.

Name	Role	Signature	Date
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